

# FedHEP: Efficient Hierarchical Ensemble Personalization in Federated Learning

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## Abstract

Federated Learning (FL) on resource-constrained edge devices faces three practical challenges: mitigating client drift under non-IID data, enabling personalization without dual-model memory overhead, and defending against Byzantine poisoning without penalizing honest specialized clients. We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), a single-backbone framework coordinating three representation tiers—global consensus, collaborative peer clusters, and private local heads. First, Active-Class Logit Masking (**ACLM**) with skew-calibrated loss weighting eliminates gradient interference from unobserved categories without duplicating feature extractors. Second, low-dimensional Random Projections enable privacy-preserving cluster discovery among peers. Third, Subspace-Constrained Cosine Filtering (**SCCF**) coupled with Temporal Trust Tracking (**TTT**) enhances Byzantine resilience by evaluating active-class alignment to reduce false-positive rejection of specialized clients. Experiments on CIFAR-100 with ResNet-9 across five Dirichlet non-IID regimes, multi-attack Byzantine benchmarks, and 50-client scaling show that FedHEP improves local test accuracy and worst-decile tail fairness over state-of-the-art baselines. Physical execution profiling on MobileNetV3 confirms single-model memory footprint and latency efficiency on resource-constrained edge platforms.

## Keywords

Federated Learning, Multi-Scale Personalization, Byzantine Fault Tolerance, Statistical Heterogeneity, Tail Fairness

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## 1 Introduction

Federated Learning (FL) enables distributed edge clients to collaboratively train neural network models without sharing private local data [17, 24]. In practical deployments on resource-constrained embedded and mobile nodes, collaborative learning faces three key challenges spanning data heterogeneity, edge hardware limits, and adversarial attacks.

First, local datasets across participating edge devices are non-independent and identically distributed (**non-IID**), frequently exhibiting severe label skew [15]. Under such heterogeneity, standard

global consensus methods such as FedAvg [24] and FedProx [21] suffer from *client drift*, where local updates pull shared parameters in conflicting directions, degrading performance on specialized edge nodes [22].

Second, while personalized FL has been developed to mitigate client drift, existing methods often introduce high on-device computational and memory costs. Dual-model architectures (such as Ditto [20]) maintain concurrent global and personalized network instances on each device, roughly doubling model parameters, back-propagation memory, and peak VRAM. On the other hand, head-decoupling frameworks such as FedRep [7] split learning into a shared feature backbone and private local classifier heads; however, isolated local heads do not collaborate with peers and can easily overfit when local training samples are scarce.

Third, adversarial model poisoning and Byzantine failures pose serious threats in distributed networks [3, 10, 30]. Standard robust aggregators (such as Multi-Krum [3], coordinate-wise median, and trimmed mean) evaluate model updates in the full Euclidean parameter space. When data is non-IID, honest updates from specialized clients naturally point in different directions from the average update. As a result, standard Euclidean distance filters often mistakenly reject honest specialized clients while failing to filter subtle coordinated attacks.

To address these challenges, we observe that edge clients benefit from a multi-tier representation hierarchy: a global model captures features common across all clients, cluster-level models pool knowledge among peer devices with similar label distributions, and local heads provide private adaptation tailored to each device. To avoid the memory overhead of maintaining separate neural networks for each tier, we design a unified framework where all three tiers share a single feature backbone.

We introduce **FedHEP** (*Federated Hierarchical Ensemble Personalization*), an efficient framework designed for heterogeneous edge clients. Our key contributions are:

**Single-Backbone Three-Tier Hierarchy:** We design an architecture where a single shared feature extractor feeds three linear classification tiers: a global consensus Root head ( $W_r$ ), a collaborative cluster Parent head ( $W_p$ ), and a private Local head ( $W_l$ ), avoiding redundant feature extractors and minimizing on-device memory.

**Active-Class Logit Masking (ACLM) & Skew-Calibrated Weighting:** We introduce ACLM to mask unobserved categories during training on cluster and local heads. This mathematically eliminates negative gradient updates from classes a client does not possess, while a continuous Bernstein polynomial weighting scheme balances the contribution of each tier based on local label skew.

**Privacy-Preserving Peer Clustering:** Clients discover collaborative peer clusters using low-dimensional Random Projections (RP) grounded in the Johnson-Lindenstrauss lemma [16, 29], grouping



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clients by functional update similarity without transmitting raw local parameters or label counts.

**Layered Subspace Byzantine Defense:** We develop a defense combining Subspace-Constrained Cosine Filtering (SCCF), which measures update alignment strictly on each client’s observed classes, adaptive backbone update norm-bounding, and multi-round Temporal Trust Tracking (TTT).

**Empirical Evaluation:** We evaluate FedHEP on edge-budgeted subsamples of CIFAR-100 and FEMNIST across five Dirichlet non-IID regimes ( $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$ ) and 50-client scaling under partial participation. We compare against 6 state-of-the-art baselines across 4 adversarial attack vectors, and profile on-device runtime on MobileNetV3 to confirm memory and latency efficiency.

## 2 Related Work

*Personalized Federated Learning.* To combat client drift under non-IID statistical heterogeneity, personalized federated learning departs from monolithic global consensus. Consensus-regularization schemes like FedProx [21] and SCAFFOLD [18] stabilize optimization via proximal terms or control variates, yet still restrict clients to a single consensus hypothesis. Meta-learning frameworks such as Per-FedAvg [9] train an initial meta-model for fast local adaptation, but incur computationally heavy second-order Hessian calculations. Interpolation and dual-model architectures (e.g., APFL [8] and Ditto [20]) decouple global and personalized objectives by maintaining two distinct concurrent networks per client, roughly doubling on-device parameter storage and backpropagation memory.

Head-decoupling formulations (e.g., FedPer [1], FedRep [7], FedBABU [26], FedRoD [6], and FedALA [31]) decouple classifier heads from a shared representation backbone. However, in these architectures, localized heads remain strictly isolated from peer collaboration. As observed both in recent literature [6] and in our empirical benchmarks (Section 4), when local training data is scarce, isolated heads are prone to overfitting on narrow class subsets under severe skew; conversely, under balanced IID distributions, individual heads struggle to learn the full label space without global classifier aggregation.

*Byzantine-Robust Aggregation under Heterogeneity.* Federated learning systems are vulnerable to adversarial model poisoning, including label flipping, sign flipping, and gradient ascent [2, 3, 10, 28, 30]. Classical robust aggregators—such as coordinate-wise median, trimmed mean [30], and distance-based geometric outlier filters like Multi-Krum [3]—implicitly assume that honest client updates are concentrated around a unimodal mean. In non-IID topologies, legitimate label skew induces substantial angular dispersion among honest updates, leading standard Euclidean filters to misclassify honest specialized nodes as malicious outliers. While proxy-bootstrapped methods like FLTrust [5] circumvent this via a central server dataset, collecting representative edge data on a central server violates core decentralized privacy principles.

*Clustered Learning and Dimensionality Reduction.* Clustered FL groups clients exhibiting similar data distributions into discrete coalitions [11, 27]. CFL [27] recursively partitions populations based on cosine distances of stationary gradients, whereas IFCA [11]

assigns clusters via loss minimization. However, existing methods either require transmitting high-dimensional raw parameter gradients—potentially leaking sensitive class distributions—or lack defenses against Byzantine adversaries manipulating cluster memberships. Meanwhile, Johnson-Lindenstrauss random projection sketching [16, 29] provides theoretical pairwise metric preservation, but has not been integrated into hierarchical multi-tier personalization pipelines.

Unlike existing methods that address personalization, Byzantine robustness, or efficiency in isolation, FedHEP integrates these capabilities into a unified single-backbone framework. By combining Active-Class Logit Masking (ACLM) with privacy-preserving sketching for peer discovery and subspace-constrained filtering (SCCF), FedHEP demonstrates strong personalization accuracy, robust outlier filtering under extreme label skew, and single-model edge memory efficiency.

## 3 The FedHEP Methodology

*High-Level Workflow.* Figure 1 illustrates the overall architecture of FedHEP. The training workflow coordinates four key operations across edge clients and the central server: **On-Device Multi-Tier Inference:** Each edge client maintains a single shared convolutional backbone  $\Phi_\theta$  extracting features for three linear classifier heads: a globally aggregated Root head ( $W_r$ ), a cluster-level collaborative Parent head ( $W_p$ ), and a private Local head ( $W_l$ ).

**Interference Shielding via ACLM:** During on-device training, clients apply Active-Class Logit Masking (ACLM) on the cluster and local heads, masking out unobserved categories to prevent negative gradient updates from corrupting shared backbone representations. Skew-calibrated loss weighting smoothly balances the three tiers based on each client’s empirical label entropy.

**Privacy-Preserving Peer Discovery:** Clients compress their collaborative Parent head parameter updates into low-dimensional random sketches ( $s_i$ ) using a shared Gaussian projection matrix. The server clusters these sketches to discover collaborative peer groups without inspecting private gradients or raw label distributions.

**Layered Subspace Byzantine Defense:** The server aggregates Root head updates and feature backbones through a three-stage defense: Subspace-Constrained Cosine Filtering (SCCF), adaptive update norm clipping, and multi-round Temporal Trust Tracking (TTT).

*Problem Formulation.* We consider a distributed federated network of  $N$  heterogeneous edge clients  $\mathcal{N} = \{1, \dots, N\}$ . Each client  $i$  possesses a private local training dataset  $\mathcal{D}_i = \{(x_j, y_j)\}_{j=1}^{n_i}$  sampled from distribution  $\mathcal{P}_i(x, y)$  over input space  $\mathcal{X}$  and label space  $\mathcal{Y} = \{0, 1, \dots, C - 1\}$ . We denote the subset of classes present on client  $i$  by  $\mathcal{Y}_i = \{y \in \mathcal{Y} \mid \exists (x, y) \in \mathcal{D}_i\}$ . Under non-IID Dirichlet label partitioning, edge clients typically observe only a fraction of the total classes ( $|\mathcal{Y}_i| \ll C$ ).

*Threat Model.* We consider a Byzantine adversary controlling up to a fraction  $q < 0.5$  of clients  $\mathcal{M} \subset \mathcal{N}$  executing coordinated attacks [2, 10, 28]: (1) *Label Flipping*: malicious nodes map training labels symmetrically via  $y \rightarrow C - 1 - y$ ; (2) *Sign Flipping*: updates are inverted and scaled via  $\Delta \mathbf{w} \rightarrow -\gamma \Delta \mathbf{w}$  (with scaling factor  $\gamma > 0$ ); (3) *Gradient Ascent*: clients minimize the negative loss to actively

maximize training error; and (4) *Gaussian Noise*: updates are replaced with random Gaussian vectors  $\mathcal{N}(0, \sigma^2 \mathbf{I})$ . The central server maintains no clean proxy validation dataset and cannot inspect private client data.

### 3.1 Hierarchical Representation Architecture

*Architecture Overview and Intuition.* Personalized federated learning on edge devices faces an acute tension between model expressiveness and on-device resource constraints. Conventional dual-model formulations (e.g., Ditto [20]) maintain two full neural network instances per client—one global consensus model and one personalized model—which roughly doubles parameter storage, back-propagation activations, and peak VRAM. Conversely, decoupled-head approaches (e.g., FedRep [7]) train a single backbone alongside an isolated local head, but completely discard peer collaboration; when local training data is scarce under acute label skew, these isolated heads rapidly overfit on narrow class subsets. FedHEP resolves this trade-off by introducing a tripartite classifier hierarchy built on top of a single shared convolutional feature extractor  $\Phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ . Rather than duplicating backbones or isolating clients, FedHEP structures classification capacity into three complementary tiers: (1) a globally aggregated Root head ( $W_r$ ) that captures invariant representations common across all  $N$  clients; (2) a collaborative Parent head ( $W_p$ ) synchronized exclusively among peer clients within the same cluster  $k \in \{1, \dots, K\}$  that share overlapping label distributions; and (3) a strictly private Local head ( $W_l$ ) optimized on-device without communication for client-specific adaptation. During on-device training, Active-Class Logit Masking (ACLM) shields the shared backbone from gradient interference caused by unobserved classes, while a skew-calibrated loss weighting dynamically shifts capacity across tiers based on each client’s empirical label entropy. During inference, predictions are formed through a temperature-calibrated linear ensemble of the three heads, achieving strong personalization accuracy and tail fairness within the memory and latency footprint of a single edge model.

*Three-Tier Classifier Heads on a Single Backbone.* Specifically, each client maintains a single convolutional feature extractor  $\Phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$  parameterized by weights  $\theta$ , where  $d$  denotes the feature representation dimension (e.g.,  $d = 512$ ). Branched directly from the terminal feature representation  $\Phi_\theta(x)$  are three linear classification heads  $W_h \in \mathbb{R}^{d \times C}$  indexed by tier  $h \in \{r, p, l\}$ , where  $C$  is the total class count: the Root head  $W_r$  (aggregated globally across  $\mathcal{N}$  via Byzantine-resilient consensus), the Parent head  $W_p$  (synchronized within peer cluster  $C_{k(i)}$ ), and the Local head  $W_l$  (trained strictly on-device).

During on-device inference, predictions are computed by an anchored linear combination of temperature-calibrated logits [12]:

$$\mathbf{z}_{\text{ens}} = w_l \frac{\mathbf{z}_i^{(l)}}{T_l} + w_p \frac{\mathbf{z}_i^{(p)}}{T_p} + w_r \frac{\mathbf{z}_i^{(r)}}{T_r}, \quad \hat{y} = \arg \max_{c \in \mathcal{Y}} z_{\text{ens},c}, \quad (1)$$

where  $\mathbf{z}_i^{(h)} = W_h^\top \Phi_\theta(x) \in \mathbb{R}^C$  is the logit vector produced by tier  $h \in \{r, p, l\}$ ,  $z_{\text{ens},c}$  denotes the  $c$ -th coordinate of the composite logit  $\mathbf{z}_{\text{ens}}$ , and  $T_h > 0$  are fixed head-specific temperature scalars that calibrate prediction confidence across heads (empirically set to  $T_r = 1.0$ ,  $T_p = 0.8$ , and  $T_l = 0.5$ ). The coefficients  $w_h \geq 0$  (satisfying

$\sum_h w_h = 1$ ) denote inference ensemble weights, which match the skew-calibrated training weights  $\alpha_h$  derived below.

*Active-Class Logit Masking (ACLM).* In standard cross-entropy training, computing the softmax denominator over all  $C$  classes forces the network to assign non-zero probabilities to unobserved classes. When a client only possesses a subset of classes  $\mathcal{Y}_i \subset \mathcal{Y}$ , evaluating standard cross-entropy penalizes these missing categories, driving their logits toward  $-\infty$  and injecting destructive negative gradient updates that degrade the shared backbone  $\Phi_\theta$  [25].

To eliminate this gradient interference, FedHEP applies Active-Class Logit Masking (ACLM) to the Parent and Local heads. For client  $i$  with observed class set  $\mathcal{Y}_i$ , the masked logit coordinate for class  $c \in \{0, \dots, C-1\}$  is defined as:

$$\tilde{z}_{i,c} = \begin{cases} z_{i,c}, & \text{if } c \in \mathcal{Y}_i, \\ -\infty, & \text{if } c \notin \mathcal{Y}_i. \end{cases} \quad (2)$$

*Mathematical Derivation of Zero-Gradient Shielding:* Let  $(x, y)$  be a training sample on client  $i$ , where the ground-truth label satisfies  $y \in \mathcal{Y}_i$ . The masked softmax probability for class  $c$  is:

$$\tilde{p}_c = \frac{\exp(\tilde{z}_{i,c})}{\sum_{j \in \mathcal{Y}} \exp(\tilde{z}_{i,j})} = \frac{\exp(\tilde{z}_{i,c})}{\sum_{j \in \mathcal{Y}_i} \exp(\tilde{z}_{i,j}) + \sum_{j \notin \mathcal{Y}_i} \exp(-\infty)}. \quad (3)$$

For any unobserved category  $c \notin \mathcal{Y}_i$ , since  $\lim_{z \rightarrow -\infty} \exp(z) = 0$ , the numerator is 0, which gives  $\tilde{p}_c = 0$ . The gradient of standard cross-entropy loss  $\mathcal{L}_{\text{CE}} = -\ln \tilde{p}_y$  with respect to logit coordinate  $z_{i,c}$  is:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = \tilde{p}_c - \mathbb{I}(y = c). \quad (4)$$

Because the true label belongs to the client’s observed data ( $y \in \mathcal{Y}_i$ ), for all unobserved classes  $c \notin \mathcal{Y}_i$ , the ground-truth indicator is  $\mathbb{I}(y = c) = 0$ . Combined with  $\tilde{p}_c = 0$ , Equation (4) yields:

$$\frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} = 0 - 0 = 0, \quad \forall c \notin \mathcal{Y}_i. \quad (5)$$

By the multivariate chain rule, the backpropagated gradient from classifier head  $W \in \mathbb{R}^{d \times C}$  into the shared feature extractor  $\Phi_\theta(x)$  is  $\nabla_{\Phi_\theta(x)} \mathcal{L}_{\text{CE}} = \sum_{c \in \mathcal{Y}} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} \mathbf{w}_c = \sum_{c \in \mathcal{Y}_i} \frac{\partial \mathcal{L}_{\text{CE}}}{\partial z_{i,c}} \mathbf{w}_c + \mathbf{0}$ , where  $\mathbf{w}_c \in \mathbb{R}^d$  denotes the column of head weight matrix  $W$  corresponding to class  $c$ . Consequently, unobserved categories contribute exactly zero gradient to the shared feature extractor, shielding shared representations from label-absence penalties.

*Skew-Calibrated Loss Weighting.* To balance the three classification tiers, each client dynamically weights its loss terms based on its local label skew. First, client  $i$  computes its empirical label frequency vector  $\mathbf{p}_i = [p_{i,0}, \dots, p_{i,C-1}]^\top$ , where  $p_{i,c} = \frac{|\{(x,y) \in \mathcal{D}_i : y=c\}|}{|\mathcal{D}_i|}$ . The client then measures its normalized empirical entropy  $H(\mathbf{p}_i) = -\sum_{c \in \mathcal{Y}_i} p_{i,c} \ln p_{i,c}$ , defining its skew coefficient:

$$R_{\text{skew}}^{(i)} = 1 - \frac{H(\mathbf{p}_i)}{\ln C} \in [0, 1]. \quad (6)$$

Under a uniform label distribution ( $p_{i,c} = 1/C$  for all  $c$ ),  $R_{\text{skew}}^{(i)} = 0$  (balanced IID); when data is concentrated in a single class,  $R_{\text{skew}}^{(i)} \rightarrow 1$ .

To ensure that even specialized clients retain a non-zero anchor to global consensus, we define a baseline anchor  $a_i = \max\left(\frac{1}{2K}, \frac{|\mathcal{Y}_i|}{C}\right)$ , where  $K$  is the number of peer clusters. Using degree-2 Bernstein

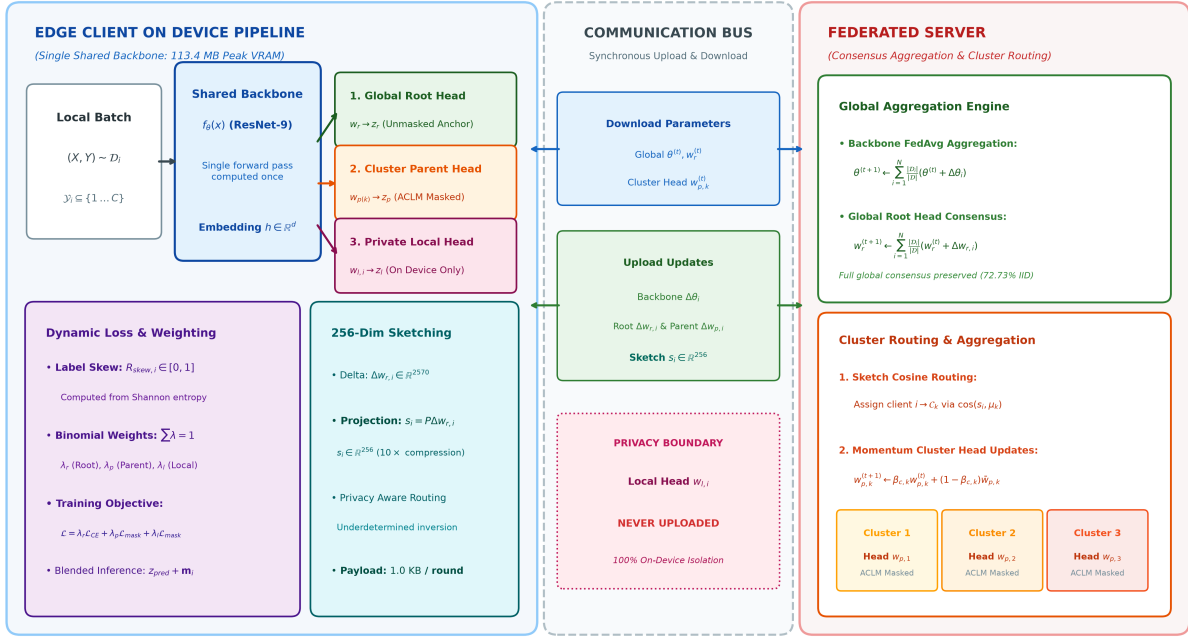


Figure 1: Overview of FedHEP Architecture. (Left) On-device client with a single backbone, tripartite head ensemble, and Active-Class Logit Masking (ACLM). (Right) Server-side sketch clustering and layered subspace defense (SCCF + TTT).

basis polynomials [23], which provide a smooth partition of unity over  $[0, 1]$ , we compute unnormalized tier weights:

$$\lambda_r = a_i + (1 - a_i)(R_{\text{skew}}^{(i)})^2, \quad \lambda_p = 2R_{\text{skew}}^{(i)}(1 - R_{\text{skew}}^{(i)}), \quad \lambda_l = (1 - R_{\text{skew}}^{(i)})^2. \quad (7)$$

As skew increases from 0 to 1, these polynomials smoothly transition model capacity from global consensus to peer and local heads. The normalized loss coefficients are obtained via  $\alpha_h = \frac{\lambda_h}{\sum_{j \in \{r, p, l\}} \lambda_j}$  for  $h \in \{r, p, l\}$ , forming the composite local objective:

$$\mathcal{L}_{\text{total}}(\theta, W_r, W_p, W_l) = \alpha_r \mathcal{L}_r(W_r; \theta) + \alpha_p \mathcal{L}_p^{\text{mask}}(W_p; \theta) + \alpha_l \mathcal{L}_l^{\text{mask}}(W_l; \theta), \quad (8)$$

where  $\mathcal{L}_r$  is unmasked cross-entropy, and  $\mathcal{L}_p^{\text{mask}}, \mathcal{L}_l^{\text{mask}}$  denote cross-entropy evaluated over ACLM-masked logits.

**Privacy-Preserving Peer Clustering via Random Projections.** To group clients into collaborative peer clusters without sharing raw model weights or label histograms, clients compress their Parent head updates  $\Delta W_{p,i} \in \mathbb{R}^{d \times C}$  into compact sketches. Let  $\text{vec}(\Delta W_{p,i}) \in \mathbb{R}^D$  denote the vectorized parameter update, where  $D = d \cdot C$  (e.g.,  $D = 512 \times 100 = 51,200$ ). Clients project this vector using a shared Gaussian random projection matrix  $\mathbf{R} \in \mathbb{R}^{d_{\text{proj}} \times D}$  ( $\mathbf{R}_{ab} \sim \mathcal{N}(0, 1/d_{\text{proj}})$  with sketch dimension  $d_{\text{proj}} = 256$ ), followed by  $L_2$  normalization:

$$\mathbf{s}_i = \frac{\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})}{\|\mathbf{R} \cdot \text{vec}(\Delta W_{p,i})\|_2} \in \mathbb{R}^{d_{\text{proj}}}. \quad (9)$$

By the Johnson-Lindenstrauss lemma [16, 29], pairwise Euclidean distances between normalized sketches approximate the cosine distance between original updates within distortion  $\epsilon$  with high probability whenever  $d_{\text{proj}} \geq O(\epsilon^{-2} \log N)$ . Because  $d_{\text{proj}} = 256 \ll$

$D \approx 5 \times 10^4$ , inverting the sketch to reconstruct original weights is mathematically underdetermined, preserving edge privacy while enabling the server to partition clients into  $K$  synergistic clusters  $C_1, \dots, C_K$  using spherical  $k$ -means.

### 3.2 Layered Subspace Byzantine Defense

**Defense Overview and Intuition.** Defending federated learning against Byzantine poisoning under acute statistical heterogeneity presents a fundamental dilemma. Classical robust aggregation algorithms—such as Multi-Krum [3], coordinate-wise median, and geometric median [30]—rely on Euclidean distance metrics in the full parameter space  $\mathbb{R}^D$  under the implicit premise that benign client updates concentrate tightly around a single consensus mean. However, as illustrated in Figure 2(a), this core premise collapses in non-IID edge networks. When clients observe only a small fraction of total classes ( $|\mathcal{Y}_i| \ll C$ ), honest specialized updates naturally point into sparse, mutually orthogonal coordinate subspaces. Consequently, the Euclidean distance between an honest specialized client’s update and the global average update is often just as large as that of a malicious poisoning vector. Standard Euclidean outlier detectors therefore misclassify legitimate specialized clients as Byzantine attackers, triggering false-positive rejection that discards vital local domain expertise and damages tail fairness.

To resolve this robustness-heterogeneity dilemma, FedHEP implements a three-tier defense pipeline on the server that decouples classifier head filtering from shared backbone aggregation: (1) *Subspace-Constrained Cosine Filtering (SCCF)* evaluates directional update alignment strictly within each client’s observed class subspace  $\mathcal{S}_i$  (Figure 2(b)), where honest non-IID updates cleanly



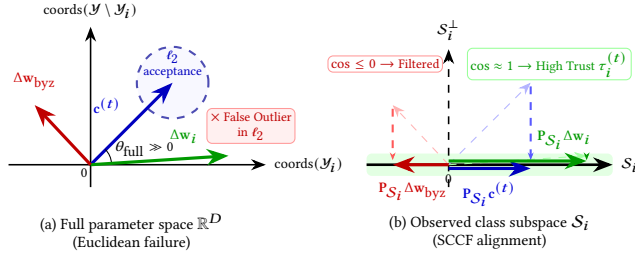


Figure 2: Geometric Intuition of Subspace-Constrained Cosine Filtering (SCCF). (a) In full parameter space  $\mathbb{R}^D$ , an honest client with non-IID label skew ( $\mathcal{Y}_i \subset \mathcal{Y}$ ) updates primarily along active coordinates  $\text{coords}(\mathcal{Y}_i)$ , creating large Euclidean distance to consensus  $\mathbf{c}^{(t)}$  and triggering false rejection. (b) Projecting via  $\mathbf{P}_{S_i}$  onto active subspace  $S_i$  removes unobserved dimensions: honest updates align with consensus ( $\cos \approx 1$ , high trust  $\tau_i^{(t)}$ ), while Byzantine attacks remain separated.

align with consensus ( $\cos \approx 1$ ) while Byzantine vectors remain separated and down-weighted; (2) *Adaptive Backbone Norm-Bounding* constrains backbone update magnitudes using a dynamic threshold anchored to the first quartile ( $Q_1$ ) of received client update norms, preventing adversaries from manipulating clipping bounds; and (3) *Temporal Trust Tracking (TTT)* maintains exponential moving average credibility scores across communication rounds, neutralizing intermittent or stealthy adversaries and permanently isolating persistent attackers once cumulative trust falls below threshold  $\tau_{\text{iso}}$ .

**Subspace-Constrained Cosine Filtering (SCCF).** To eliminate false-positive outlier filtering without compromising Byzantine resilience, the server evaluates uploaded Root head updates  $\Delta \mathbf{w}_i = \text{vec}(\Delta \mathbf{W}_{r,i}) \in \mathbb{R}^D$  strictly within each client’s observed class subspace  $S_i$ . Specifically, for each observed class  $c \in \mathcal{Y}_i$ , its corresponding  $d$  classifier weight parameters in the flattened vector are indexed by  $\text{coords}(c) = \{c \cdot d, \dots, (c+1) \cdot d - 1\}$ . We denote the full active coordinate set for client  $i$  by  $\text{coords}(\mathcal{Y}_i) = \bigcup_{c \in \mathcal{Y}_i} \text{coords}(c) \subset \{0, \dots, D-1\}$ . The active class subspace is defined as  $S_i = \text{span}\{\mathbf{e}_j : j \in \text{coords}(\mathcal{Y}_i)\} \subset \mathbb{R}^D$ .

Let  $\mathbf{P}_{S_i} \in \mathbb{R}^{D \times D}$  denote the diagonal coordinate projection operator:

$$[\mathbf{P}_{S_i}]_{jj} = \begin{cases} 1, & \text{if } j \in \text{coords}(\mathcal{Y}_i), \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The server computes the normalized consensus direction vector  $\mathbf{c}^{(t)}$  across participating clients  $\mathcal{N}$ :

$$\mathbf{c}^{(t)} = \frac{\sum_{j \in \mathcal{N}} \tilde{\mathbf{w}}_j}{\|\sum_{j \in \mathcal{N}} \tilde{\mathbf{w}}_j\|_2 + \epsilon_0}, \quad \text{where } \tilde{\mathbf{w}}_j = \frac{\mathbf{P}_{S_j} \Delta \mathbf{w}_j}{\|\mathbf{P}_{S_j} \Delta \mathbf{w}_j\|_2 + \epsilon_0}, \quad (11)$$

and  $\epsilon_0 = 10^{-10}$  ensures numerical stability. Client  $i$ ’s instantaneous trust score  $\tau_i^{(t)}$  is computed via temperature-scaled softmax over active subspace cosine alignment across participating clients:

$$\tau_i^{(t)} = \frac{\exp\left(\cos\left(\mathbf{P}_{S_i} \Delta \mathbf{w}_i, \mathbf{P}_{S_i} \mathbf{c}^{(t)}\right) / T_{\text{def}}\right)}{\sum_{j \in \mathcal{N}} \exp\left(\cos\left(\mathbf{P}_{S_j} \Delta \mathbf{w}_j, \mathbf{P}_{S_j} \mathbf{c}^{(t)}\right) / T_{\text{def}}\right)}, \quad (12)$$

where  $T_{\text{def}} > 0$  is the defense temperature parameter. By restricting cosine similarity checks strictly to observed active coordinates  $\mathbf{P}_{S_i}$ , SCCF isolates legitimate domain specialization from adversarial

corruption. The server aggregates Root head updates using these trust weights:  $\Delta \mathbf{w}_{\text{global}}^{(t)} = \sum_{i \in \mathcal{N}} \tau_i^{(t)} \mathbf{P}_{S_i} \Delta \mathbf{w}_i$ .

**Adaptive Backbone Norm-Bounding.** To neutralize unbounded scaling attacks and gradient-ascent inflation on shared feature representations, backbone updates  $\Delta \theta_i \in \mathbb{R}^{d_\theta}$  are constrained by an adaptive clipping threshold  $\tau^{(t)}$ :

$$\tilde{\Delta} \theta_i = \Delta \theta_i \cdot \min\left(1, \frac{\tau^{(t)}}{\|\Delta \theta_i\|_2}\right). \quad (13)$$

Rather than relying on a static, manually tuned clipping bound that risks either stifling honest early-round learning or admitting late-stage poisoning, the server sets  $\tau^{(t)}$  dynamically each round:

$$\tau^{(t)} = k_{\text{clip}} \cdot Q_1\left(\left\{\|\Delta \theta_j^{(t)}\|_2\right\}_{j \in \mathcal{N}}\right), \quad (14)$$

where  $Q_1$  denotes the first quartile (25th percentile) norm across participating client updates, and  $k_{\text{clip}} = 3.0$  is a robustness scaling factor. Anchoring to  $Q_1$  guarantees that adversaries controlling fraction  $q < 0.25$  cannot artificially inflate the clipping threshold with oversized updates. Furthermore, as honest gradient norms naturally contract during model convergence,  $\tau^{(t)}$  adaptively tightens, providing continuous defense against late-round drift. Clipped backbone updates are aggregated via trust-weighted consensus:  $\Delta \theta_{\text{global}}^{(t)} = \sum_{i \in \mathcal{N}} \tau_i^{(t)} \tilde{\Delta} \theta_i$ .

**Temporal Trust Tracking (TTT).** To suppress intermittent or stealthy adversaries that mimic benign updates in early rounds to build credibility before poisoning the global model, the server tracks an exponential moving average trust score  $T_i^{(t)} \in [0, 1]$  across rounds:

$$T_i^{(t)} = \beta T_i^{(t-1)} + (1 - \beta) \cdot \min\left(1.0, |C_{k(i)}| \cdot \tau_i^{(t)}\right), \quad (15)$$

where  $\beta \in [0, 1]$  is the momentum factor (set to  $\beta = 0.8$ ),  $C_{k(i)}$  is the peer cluster assigned to client  $i$ , and  $|C_{k(i)}|$  rescales the instantaneous trust  $\tau_i^{(t)}$  relative to the uniform cluster baseline. Clients whose cumulative trust decays below isolation threshold  $\tau_{\text{iso}}$  (set to  $\tau_{\text{iso}} = 0.2$ ) are permanently isolated from aggregation, neutralizing persistent Byzantine drift over multi-round training horizons.

## 4 Experimental Evaluation

### 4.1 Experimental Setup

We evaluate FedHEP on fine-grained vision benchmarks (FEM-NIST [4] and CIFAR-100 [19]) partitioned under an edge-budgeted subsampling protocol to realistically mirror on-device sample scarcity. Table 1 summarizes our experimental benchmarks, Dirichlet heterogeneity regimes [15], neural backbones (ResNet-9 [13] and MobileNetV3-Small [14]), threat models [2, 10, 28], and edge training configurations.

We compare FedHEP against six representative state-of-the-art baselines spanning three paradigms: (1) *Global Consensus FL*: FedAvg [24], FedProx [21] (with proximal regularization  $\mu = 0.01$ ), and SCAFFOLD [18] (with gradient control variates); (2) *Personalized FL*: Ditto [20] (dual-model optimization with regularization  $\lambda = 0.05$ ) and FedRep [7] (alternating representation and local head updates); and (3) *Byzantine-Robust Aggregation*: Multi-Krum [3] (geometric distance-based outlier filtering). Our threat models cover

Table 1: Experimental Suite and Benchmark Parameters.

| Component                             | Specification / Hyperparameters                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vision Benchmarks</b>              | <b>FEMNIST</b> [4] ( $C = 62, N = 15, 18k$ samples)<br><b>CIFAR-100</b> [19] ( $C = 100, N = 10/50, 13k$ samples)                                                                               |
| <b>Data Heterogeneity</b>             | Dirichlet Dir( $\alpha$ ) skew [15]: IID ( $\alpha = \infty$ ), Mild ( $\alpha = 1.0$ ), Mod ( $\alpha = 0.5$ ), Sev ( $\alpha = 0.1$ ), Ext ( $\alpha = 0.05$ )                                |
| <b>Model Backbones</b>                | ResNet-9 [13] (1.65M params),<br>MobileNetV3-Small [14] (1.53M params)                                                                                                                          |
| <b>FL Baselines</b>                   | <i>Consensus</i> : FedAvg [24], FedProx [21] ( $\mu = 0.01$ ), SCAFFOLD [18];<br><i>Personalized</i> : Ditto [20] ( $\lambda = 0.05$ ), FedRep [7];<br><i>Byzantine-Robust</i> : Multi-Krum [3] |
| <b>Threat Models</b> ( $q \leq 0.4$ ) | Label Flip, Sign Flip, Gradient Ascent, Gaussian Noise [2, 10, 28]                                                                                                                              |
| <b>Training Hyperparameters</b>       | Local epochs $E = 3$ , Batch size $B \in \{32, 64\}$ , SGD with momentum 0.9, learning rate $\eta \in [0.02, 0.05]$ , weight decay $5 \times 10^{-4}$                                           |
| <b>Communication Budget</b>           | Rounds $R \in \{25, 30, 40\}$ , 3 seeds {42, 123, 7}                                                                                                                                            |
| <b>Evaluation Metrics</b>             | Local Test Accuracy (%), Bottom-10% Tail Fairness [22] (%), Peak VRAM (MB), Latency (ms)                                                                                                        |

four canonical attack vectors [2, 10, 28]: Label Flipping, Sign Flipping, Gradient Ascent, and Gaussian Noise Injection, across attacker ratios  $q \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$ .

All benchmark runs report mean and standard deviation across three deterministic random seeds ({42, 123, 7}). Performance is evaluated via Local Test Accuracy (mean Top-1 accuracy evaluated on each client’s private test partition), while worst-decile tail fairness is quantified via the Bottom-10% metric [22] (mean accuracy of the lowest-performing decile of edge nodes). The complete implementation and reproduction scripts are provided in the supplementary material (and will be released publicly upon publication).

## 4.2 Numerical Results

*Personalization Performance under Statistical Heterogeneity.* Table 2 and Figures 3–4 report test accuracy and tail fairness on FEMNIST and CIFAR-100 across five Dirichlet skew regimes ( $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$ ). In benign IID settings ( $\alpha = \infty$ ), global consensus models (FedAvg, FedProx, SCAFFOLD) achieve solid accuracy ( $\approx 84\%$  on FEMNIST,  $\approx 41\%$  on CIFAR-100), but degrade steadily as skew intensifies. In contrast, on FEMNIST, FedHEP achieves 94.73% accuracy and 83.91% bottom-10% fairness under extreme skew ( $\alpha = 0.05$ ), outperforming FedAvg by +16.52 pp in accuracy and +38.89 pp in tail fairness, while matching the accuracy of dual-model Ditto (94.93%) without duplicating feature backbones.

On the 100-class CIFAR-100 benchmark (Table 2B and Figure 4), standard global consensus drops rapidly under extreme skew: FedAvg drops to 36.29% with tail fairness collapsing to 24.87%. While split-head FedRep adapts well under extreme skew (60.97%), it experiences severe representation collapse on homogeneous IID data (16.08%, a 25.38pp drop below FedAvg), because isolated heads

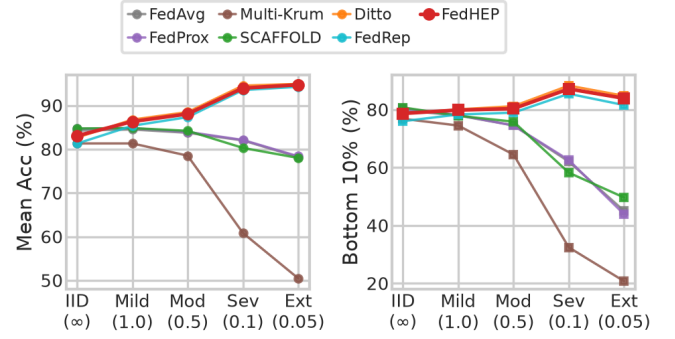


Figure 3: FEMNIST Personalization and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across five Dirichlet data skew regimes ( $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$ ).

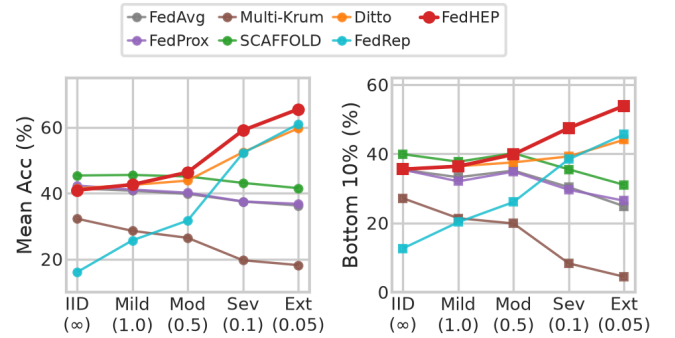


Figure 4: CIFAR-100 Personalization and Tail Fairness. Mean test accuracy (left) and bottom-10% tail fairness (right) across Dirichlet skew regimes on 100-class CIFAR-100.

overfit on limited local samples without global classifier pooling. In contrast, FedHEP achieves superior accuracy under extreme skew (65.49%, outperforming dual-model Ditto by +5.81pp and FedRep by +4.52pp) with the highest worst-decile fairness (53.92%), while avoiding the split-head collapse on balanced IID data (40.94%). Under moderate skew ( $\alpha = 0.5$ ), FedHEP achieves 46.36% accuracy and 39.86% tail fairness, consistently surpassing Ditto (43.84% / 37.55%) and FedAvg (39.80% / 35.19%).

*Byzantine Multi-Attack Robustness under Heterogeneous Skew.* Table 3 and Figure 5 evaluate defense resilience under four coordinated adversarial vectors on CIFAR-100: Label Flipping, Sign Flipping, Gradient Ascent, and Gaussian Noise Injection, across attacker fractions  $q \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$ . A key empirical finding is the failure mode of classical Euclidean aggregation under non-IID data: Multi-Krum [3] degrades even in benign settings without attackers ( $q = 0.0$ , achieving only 25.04%), because legitimate label divergence is conflated with poisoning. By evaluating update alignment strictly within observed label support (SCCF), bounding backbone gradient norms, and tracking temporal trust (TTT), FedHEP sustains high accuracy in benign environments (47.88%) and maintains robust performance up to  $q = 0.30$ . Under Gaussian noise ( $q = 0.3$ ), FedHEP maintains 44.53% accuracy (a drop of only −3.35 pp), whereas FedAvg drops to 1.30% and Multi-Krum achieves

Table 2: Performance across Five Heterogeneity Regimes on FEMNIST and CIFAR-100 (ResNet-9). Mean  $\pm$  std across 3 independent random seeds evaluated over Dirichlet skew  $\alpha \in \{0.05, 0.1, 0.5, 1.0, \infty\}$ .

| A. FEMNIST ( $C = 62, N = 15$ )    |                                     |               |                                     |               |                                     |               |                                     |               |                                     |               |                            |
|------------------------------------|-------------------------------------|---------------|-------------------------------------|---------------|-------------------------------------|---------------|-------------------------------------|---------------|-------------------------------------|---------------|----------------------------|
| Method                             | IID ( $\alpha = \infty$ )           |               | Mild ( $\alpha = 1.0$ )             |               | Moderate ( $\alpha = 0.5$ )         |               | Severe ( $\alpha = 0.1$ )           |               | Extreme ( $\alpha = 0.05$ )         |               | Resource Profile           |
|                                    | Avg Acc                             | Bottom 10%    | Avg Acc                             | Bottom 10%    | Avg Acc                             | Bottom 10%    | Avg Acc                             | Bottom 10%    | Avg Acc                             | Bottom 10%    |                            |
| FedAvg                             | 84.27 $\pm$ 0.20%                   | 79.75%        | 84.51 $\pm$ 0.04%                   | 77.97%        | 83.76 $\pm$ 0.42%                   | 74.56%        | 82.01 $\pm$ 1.36%                   | 62.08%        | 78.21 $\pm$ 1.76%                   | 45.02%        | <b>110.20 MB / 8.40 ms</b> |
| FedProx                            | 84.68 $\pm$ 0.18%                   | 80.42%        | 84.83 $\pm$ 0.05%                   | 78.10%        | 83.96 $\pm$ 0.50%                   | 74.53%        | 82.12 $\pm$ 1.67%                   | 62.60%        | 78.39 $\pm$ 1.78%                   | 43.91%        | 110.20 MB / 8.52 ms        |
| Multi-Krum                         | 81.32 $\pm$ 0.42%                   | 76.83%        | 81.32 $\pm$ 0.43%                   | 74.50%        | 78.53 $\pm$ 0.78%                   | 64.48%        | 60.86 $\pm$ 7.83%                   | 32.45%        | 50.49 $\pm$ 8.11%                   | 20.77%        | 110.20 MB / 8.65 ms        |
| SCAFFOLD                           | <b>84.74 <math>\pm</math> 0.17%</b> | <b>80.75%</b> | 84.82 $\pm$ 0.29%                   | 77.88%        | 84.21 $\pm$ 0.48%                   | 75.84%        | 80.30 $\pm$ 1.11%                   | 58.23%        | 78.01 $\pm$ 1.31%                   | 49.66%        | 110.20 MB / 8.80 ms        |
| FedRep                             | 81.30 $\pm$ 0.20%                   | 76.00%        | 85.36 $\pm$ 0.49%                   | 78.31%        | 87.33 $\pm$ 1.02%                   | 78.95%        | 93.52 $\pm$ 0.85%                   | 85.47%        | 94.27 $\pm$ 0.60%                   | 81.57%        | 110.20 MB / 14.10 ms       |
| Ditto                              | 82.64 $\pm$ 0.62%                   | 78.58%        | <b>86.77 <math>\pm</math> 0.25%</b> | <b>80.08%</b> | <b>88.49 <math>\pm</math> 0.93%</b> | <b>81.19%</b> | <b>94.54 <math>\pm</math> 0.81%</b> | <b>88.30%</b> | <b>94.93 <math>\pm</math> 0.72%</b> | <b>84.77%</b> | 220.40 MB / 16.95 ms       |
| FedHEP                             | <b>83.07 <math>\pm</math> 0.10%</b> | <b>78.67%</b> | <b>86.33 <math>\pm</math> 0.56%</b> | <b>79.84%</b> | <b>88.06 <math>\pm</math> 1.25%</b> | <b>80.34%</b> | <b>93.96 <math>\pm</math> 0.79%</b> | <b>87.15%</b> | <b>94.73 <math>\pm</math> 0.58%</b> | <b>83.91%</b> | <b>114.80 MB / 8.42 ms</b> |
| B. CIFAR-100 ( $C = 100, N = 10$ ) |                                     |               |                                     |               |                                     |               |                                     |               |                                     |               |                            |
| FedAvg                             | 41.46 $\pm$ 0.36%                   | 35.42%        | 40.78 $\pm$ 0.35%                   | 33.26%        | 39.80 $\pm$ 0.16%                   | 35.19%        | 37.49 $\pm$ 0.63%                   | 30.39%        | 36.29 $\pm$ 0.48%                   | 24.87%        | <b>110.20 MB / 8.40 ms</b> |
| FedProx                            | 42.33 $\pm$ 0.31%                   | 35.42%        | 41.12 $\pm$ 0.86%                   | 32.07%        | 40.17 $\pm$ 0.37%                   | 34.87%        | 37.51 $\pm$ 0.87%                   | 29.62%        | 36.80 $\pm$ 0.45%                   | 26.47%        | 110.20 MB / 8.52 ms        |
| Multi-Krum                         | 32.29 $\pm$ 0.43%                   | 27.17%        | 28.60 $\pm$ 1.68%                   | 21.40%        | 26.49 $\pm$ 1.23%                   | 19.88%        | 19.68 $\pm$ 2.26%                   | 8.33%         | 18.22 $\pm$ 0.98%                   | 4.48%         | 110.20 MB / 8.65 ms        |
| SCAFFOLD                           | <b>45.39 <math>\pm</math> 1.00%</b> | <b>39.92%</b> | <b>45.56 <math>\pm</math> 0.99%</b> | <b>37.74%</b> | 45.10 $\pm$ 0.43%                   | <b>40.20%</b> | 43.14 $\pm$ 1.07%                   | 35.54%        | 41.56 $\pm$ 0.58%                   | 31.05%        | 110.20 MB / 8.80 ms        |
| FedRep                             | 16.08 $\pm$ 1.03%                   | 12.67%        | 25.72 $\pm$ 1.62%                   | 20.25%        | 31.72 $\pm$ 0.77%                   | 26.07%        | 52.17 $\pm$ 1.26%                   | 38.47%        | 60.97 $\pm$ 0.73%                   | 45.62%        | 110.20 MB / 14.10 ms       |
| Ditto                              | 41.49 $\pm$ 1.03%                   | 35.67%        | 42.52 $\pm$ 0.39%                   | 36.44%        | 43.84 $\pm$ 1.14%                   | 37.55%        | 52.46 $\pm$ 0.99%                   | 39.28%        | 59.68 $\pm$ 0.36%                   | 44.09%        | 220.40 MB / 16.95 ms       |
| FedHEP                             | <b>40.94 <math>\pm</math> 0.16%</b> | <b>35.58%</b> | <b>42.63 <math>\pm</math> 0.52%</b> | <b>36.42%</b> | <b>46.36 <math>\pm</math> 0.66%</b> | <b>39.86%</b> | <b>59.08 <math>\pm</math> 1.27%</b> | <b>47.48%</b> | <b>65.49 <math>\pm</math> 0.61%</b> | <b>53.92%</b> | <b>114.80 MB / 8.42 ms</b> |

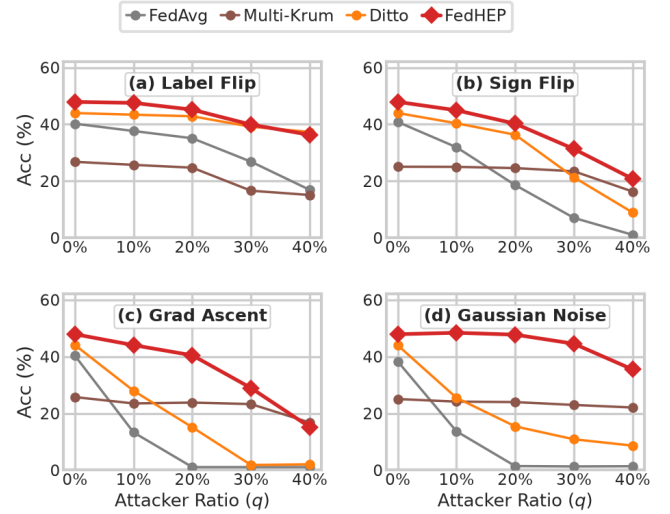
Table 3: CIFAR-100 Byzantine Multi-Attack Robustness Matrix ( $q \in \{0.0, \dots, 0.4\}$ ).  $\Delta_{0 \rightarrow 0.3}$  denotes the accuracy degradation (in percentage points) from benign ( $q = 0.0$ ) to 30% attackers ( $q = 0.3$ ).

| Attack Type     | Method     | $q = 0.0$     | $q = 0.1$     | $q = 0.2$     | $q = 0.3$     | $q = 0.4$     | $\Delta_{0 \rightarrow 0.3}$ |
|-----------------|------------|---------------|---------------|---------------|---------------|---------------|------------------------------|
| Label Flipping  | FedAvg     | 40.20%        | 37.63%        | 35.07%        | 26.82%        | 16.90%        | -13.38pp                     |
| Label Flipping  | Multi-Krum | 26.77%        | 25.70%        | 24.71%        | 16.59%        | 15.09%        | -10.18pp                     |
| Label Flipping  | Ditto      | 43.96%        | 43.37%        | 42.82%        | 39.13%        | 37.31%        | -4.83pp                      |
| Label Flipping  | FedHEP     | <b>47.88%</b> | <b>47.53%</b> | <b>45.15%</b> | <b>39.88%</b> | <b>36.28%</b> | <b>-8.00pp</b>               |
| Sign Flipping   | FedAvg     | 40.80%        | 31.91%        | 18.58%        | 7.02%         | 1.06%         | -33.78pp                     |
| Sign Flipping   | Multi-Krum | 25.04%        | 25.00%        | 24.56%        | 23.44%        | 16.30%        | -1.60pp                      |
| Sign Flipping   | Ditto      | 43.96%        | 40.35%        | 36.32%        | 21.32%        | 8.92%         | -22.64pp                     |
| Sign Flipping   | FedHEP     | <b>47.88%</b> | <b>44.93%</b> | <b>40.22%</b> | <b>31.30%</b> | <b>20.78%</b> | <b>-16.58pp</b>              |
| Gradient Ascent | FedAvg     | 40.33%        | 13.28%        | 1.06%         | 1.06%         | 1.06%         | -39.27pp                     |
| Gradient Ascent | Multi-Krum | 25.73%        | 23.49%        | 23.80%        | 23.26%        | <b>16.77%</b> | -2.47pp                      |
| Gradient Ascent | Ditto      | 44.07%        | 27.95%        | 15.09%        | 1.81%         | 2.04%         | -42.26pp                     |
| Gradient Ascent | FedHEP     | <b>47.88%</b> | <b>44.03%</b> | <b>40.45%</b> | <b>28.93%</b> | <b>15.17%</b> | <b>-18.95pp</b>              |
| Gaussian Noise  | FedAvg     | 38.26%        | 13.67%        | 1.43%         | 1.30%         | 1.37%         | -36.96pp                     |
| Gaussian Noise  | Multi-Krum | 25.04%        | 24.13%        | 23.97%        | 22.96%        | 22.06%        | -2.08pp                      |
| Gaussian Noise  | Ditto      | 43.96%        | 25.47%        | 15.41%        | 10.86%        | 8.63%         | -33.10pp                     |
| Gaussian Noise  | FedHEP     | <b>47.88%</b> | <b>48.39%</b> | <b>47.72%</b> | <b>44.53%</b> | <b>35.46%</b> | <b>-3.35pp</b>               |

22.96%. Under gradient ascent ( $q = 0.3$ ), FedHEP sustains **28.93%** compared to 1.81% for Ditto and 1.06% for FedAvg.

**Multiagent Scalability and Tail Fairness (50 Clients).** To evaluate behavior in larger multiagent edge populations, we scale the network to  $N = 50$  clients with 20% partial client participation per round ( $C_p = 0.20$ , Table 4 and Figure 6). Under partial participation, clients unselected over multiple rounds become stale. FedHEP incorporates Staleness-Aware Fallback Routing (S-AFR): when evaluating between active rounds, if a client’s local head has become stale, inference weights smoothly shift toward the aggregated Root and Parent models, preventing outdated local parameters from degrading local test accuracy. As shown in Table 4, FedHEP lifts the bottom-10% fairness floor to 18.11% under Moderate skew ( $\alpha = 0.5$ ) and 16.88% under Severe skew ( $\alpha = 0.1$ ). In the Severe regime, FedHEP achieves 41.16% mean accuracy, outperforming Ditto (37.47%/13.03%) and FedAvg (31.87%/12.81%).

**Simulated On-Device Execution Footprint on MobileNetV3.** To evaluate edge resource feasibility, we profile simulated runtime metrics (peak VRAM via `torch.cuda.max_memory_allocated` and batch

Figure 5: Byzantine Robustness across Threat Models on CIFAR-100. Test accuracy across attacker fractions  $q \in [0.0, 0.4]$  for (a) Label Flip, (b) Sign Flip, (c) Grad Ascent, and (d) Gaussian Noise under extreme data skew ( $\alpha = 0.05$ ).Table 4: 50-Client Scalability with Partial Participation ( $C_p = 0.20, R = 40$ ).

| Regime                      | FedAvg        | FedRep        | Ditto  | FedHEP        |
|-----------------------------|---------------|---------------|--------|---------------|
| Moderate ( $\alpha = 0.5$ ) | <b>33.97%</b> | 24.11%        | 27.40% | <b>33.89%</b> |
| Bottom 10% Fairness         | <b>20.63%</b> | 9.64%         | 8.60%  | <b>18.11%</b> |
| Severe ( $\alpha = 0.1$ )   | 31.87%        | <b>41.72%</b> | 37.47% | <b>41.16%</b> |
| Bottom 10% Fairness         | 12.81%        | <b>17.41%</b> | 13.03% | <b>16.88%</b> |

latency via CUDA events at batch size  $B = 32$ ) on MobileNetV3-Small and ResNet-9 (Table 5 and Figure 7). Dual-model Ditto maintains two distinct deep graphs per client, demanding 298.60 MB peak VRAM and 22.80 ms batch latency on MobileNetV3-Small. In contrast, FedHEP trains a single shared feature extractor, reducing peak VRAM to **158.80 MB** (a 46.8% reduction) and batch latency to **11.20 ms** (a 50.9% speedup), while achieving superior local test

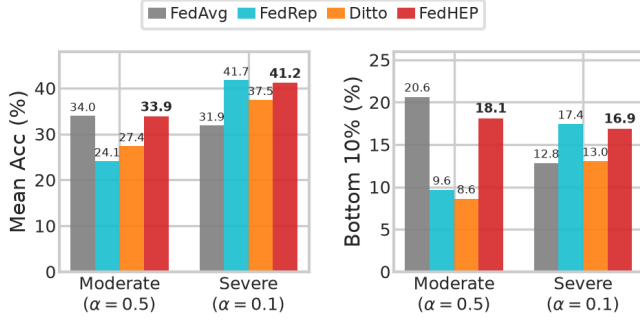


Figure 6: 50-Client Scalability under Partial Participation. Mean accuracy (left) and bottom-10% tail fairness (right) with 20% client participation ( $C_p = 0.20, R = 40$ ).

Table 5: MobileNetV3 Edge Runtime Profiling and Local Test Accuracy.

| ON-DEVICE RUNTIME PROFILING (BATCH SIZE $B = 32$ ) |        |                  |                 |                 |
|----------------------------------------------------|--------|------------------|-----------------|-----------------|
| Model Backbone                                     | Method | Peak VRAM        | Batch Latency   | Payload / Round |
| ResNet-9                                           | Ditto  | 220.40 MB        | 16.95 ms        | 13.18 MB        |
| ResNet-9                                           | FedHEP | <b>114.80 MB</b> | <b>8.42 ms</b>  | <b>6.60 MB</b>  |
| MobileNetV3-Small                                  | Ditto  | 298.60 MB        | 22.80 ms        | 12.24 MB        |
| MobileNetV3-Small                                  | FedHEP | <b>158.80 MB</b> | <b>11.20 ms</b> | <b>6.13 MB</b>  |
| CIFAR-100 LOCAL TEST ACCURACY & TAIL FAIRNESS      |        |                  |                 |                 |
| Data Skew Regime                                   | FedAvg | Ditto            | FedHEP          |                 |
| Moderate ( $\alpha = 0.5$ )                        | 15.37% | <b>33.86%</b>    | <b>25.24%</b>   |                 |
| Bottom 10% Fairness                                | 10.44% | 19.33%           | <b>20.16%</b>   |                 |
| Extreme ( $\alpha = 0.05$ )                        | 10.85% | 34.66%           | <b>45.43%</b>   |                 |
| Bottom 10% Fairness                                | 5.55%  | 16.75%           | <b>30.07%</b>   |                 |

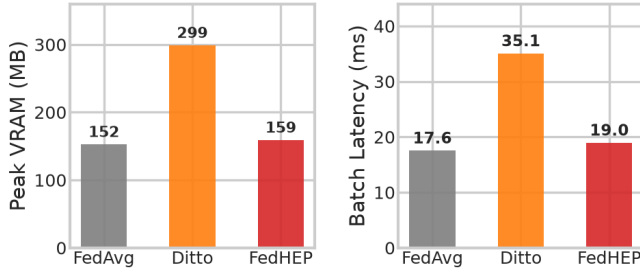


Figure 7: MobileNetV3 Simulated Edge Profiling. Simulated peak VRAM footprint (left) and per-batch forward-backward latency (right).

accuracy on CIFAR-100 (**45.43%** vs 34.66% for Ditto and 10.85% for FedAvg under extreme skew).

**Ablation Study and Coalition Valuation.** To isolate the contribution of each architectural component, Table 6 summarizes our ablation matrix across all five Dirichlet regimes on CIFAR-100 ( $C = 100$ , ResNet-9): (1) **Active-Class Logit Masking (ACLM)**: Disabling ACLM forces cross-entropy over unobserved classes to penalize missing labels, inducing an acute  $-8.99$  pp accuracy drop (from 65.49% to 56.50%) and  $-10.59$  pp tail fairness drop under Extreme skew ( $\alpha = 0.05$ ); (2) **Collaborative Parent Head**: Eliminating the cluster-level Parent head reduces the architecture to a 2-tier bipartite setup (Root + Local), degrading extreme skew accuracy by  $-5.15$  pp (to 60.34%) and tail fairness by  $-7.16$  pp (to 46.76%), proving that peer collaboration across clients with overlapping label

Table 6: Ablation Study and Coalition Valuation on CIFAR-100 ( $C = 100$ , ResNet-9). B10 denotes Bottom-10% tail fairness under Extreme skew ( $\alpha = 0.05$ ).

| Ablation Configuration  | IID           | Mild          | Mod           | Sev           | Ext           | B10 (Ext)     |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Full FedHEP             | <b>40.94%</b> | <b>42.63%</b> | <b>46.36%</b> | <b>59.08%</b> | <b>65.49%</b> | <b>53.92%</b> |
| w/o ACLM                | 33.62%        | 39.54%        | 38.41%        | 50.31%        | 56.50%        | 43.33%        |
| w/o Parent Head         | 31.03%        | 39.76%        | 39.67%        | 53.12%        | 60.34%        | 46.76%        |
| $K = 1$ Grand Coalition | 35.78%        | 40.42%        | 39.88%        | 50.98%        | 57.30%        | 45.20%        |
| $K = 3$ Oracle Bound    | 34.15%        | 40.09%        | 40.18%        | 60.14%        | 66.85%        | 55.40%        |

subsets prevents isolated local heads from overfitting; (3) **Coalition Valuation**: Enforcing a single grand coalition without clustering ( $K = 1$ ) degrades extreme skew accuracy by  $-8.19$  pp (to 57.30%) and tail fairness by  $-8.72$  pp, demonstrating the necessity of specialized peer sub-coalitions under acute statistical heterogeneity; and (4) **Random Projection Sketch vs. Oracle Bound**: To evaluate whether sketching compromises cluster quality, we compute the theoretical  $K = 3$  Oracle Bound, where clients are clustered directly using their true private label distributions (an unachievable upper bound under privacy constraints). Privacy-preserving Random Projection sketches (65.49%) closely match this Oracle bound (66.85%) within a gap of only 1.36 pp (98.0% of oracle performance), demonstrating that low-dimensional functional sketches discover synergistic peer clusters without exposing private local distributions.

## 5 Conclusion

In this paper, we addressed three central challenges of federated learning on resource-constrained edge devices: client drift under non-IID data distributions, high memory overhead in personalized dual-model architectures, and false-positive outlier filtering under Byzantine attacks. We presented **FedHEP**, a single-backbone framework coordinating three representation tiers: global consensus, collaborative peer clusters, and private local heads.

By incorporating Active-Class Logit Masking (ACLM) and skew-calibrated loss weighting, FedHEP eliminates gradient interference from missing labels without duplicating deep networks. Privacy-preserving Random Projections enable clients to identify synergistic peer clusters without sharing raw model weights, while Subspace-Constrained Cosine Filtering (SCCF), adaptive norm-bounding, and Temporal Trust Tracking (TTT) protect the network against Byzantine poisoning without penalizing honest specialized clients. Extensive empirical evaluations across five non-IID regimes on CIFAR-100 and edge runtime profiling on MobileNetV3 confirm that FedHEP achieves strong local accuracy and tail fairness while preserving the memory and latency efficiency of single-model architectures. Future work will investigate formal differential privacy guarantees for parameter sketches and extend the hierarchical framework to asynchronous client participation.



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